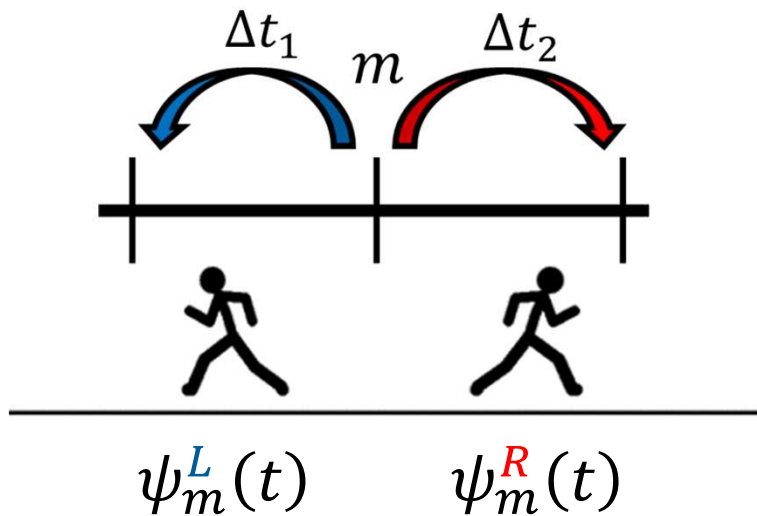


Basic concept of CTRW model



Capital letter: competition

Small letter: w/o competition

$$\psi_m^{L(R)}(t) :$$

Waiting time distribution that the jump from site m to left(right) side occurs **in absence of jump to right(left) side.**

$$\Psi_m^{L(R)}(t) = \psi_m^{L(R)}(t) \int_t^\infty d\tau \psi_m^{R(L)}(\tau) :$$

Waiting time distribution that the jump from site m to left(right) side occurs.

$\Psi_m(t) = \Psi_m^L(t) + \Psi_m^R(t)$: Normalized waiting time distribution that the jump from site m occurs.

$\Psi_m(t) = \Psi(t)$: Assume that waiting time distribution is the **same for all sites** except for the boundaries.

Basic concept of CTRW model

$f_n(t|m)dt$:

Probability that the random walker starts at a site m at time 0 and arrives at a site n between time $t \sim t + dt$

$$f_n(t|m) = \left(\int_0^t d\tau \Psi^R(\tau) f_{n-1}(t-\tau|m) + \int_0^t d\tau \Psi^L(\tau) f_{n+1}(t-\tau|m) \right) + \delta_{nm} \delta(t)$$

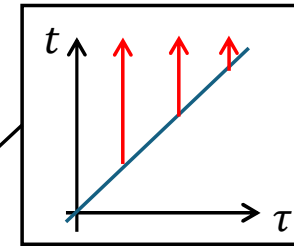
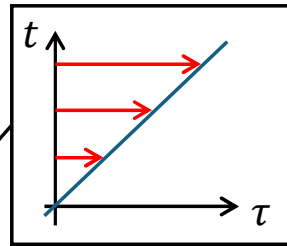
$$\hat{f}_n(u|m) \equiv \int_0^\infty dt e^{-ut} f_n(t|m) = \left(\hat{\Psi}^R(u) \hat{f}_{n-1}(u|m) + \hat{\Psi}^L(u) \hat{f}_{n+1}(u|m) \right) + \delta_{nm}$$

$$= \hat{\Psi}(u) \left(R \hat{f}_{n-1}(u|m) + L \hat{f}_{n+1}(u|m) \right) + \delta_{nm}$$

$\psi^L(t)$ 와 $\psi^R(t)$ 가
같은 distribution을 따른다면.

$$\hat{\Psi}^L(0) = L \quad \& \quad \hat{\Psi}^R(0) = R$$

$$\int_0^\infty dt e^{-ut} \left[\int_0^t d\tau a(t-\tau) b(\tau) \right] = \hat{a}(u) \hat{b}(u)$$



$$\int_0^\infty dt e^{-u} \left[\int_0^t d\tau a(t-\tau) b(\tau) \right] = \int_0^\infty dt \int_0^t d\tau [e^{-ut} a(t-\tau) b(\tau)] = \int_0^\infty d\tau \int_\tau^\infty dt [e^{-ut} a(t-\tau) b(\tau)]$$

$$= \int_0^\infty d\tau b(\tau) \int_\tau^\infty dt [e^{-ut} a(t-\tau)] = \int_0^\infty d\tau b(\tau) \int_0^\infty d\tau [e^{-u(\tau+\tau)} a(\tau)] = \int_0^\infty d\tau e^{-u\tau} b(\tau) \times \hat{a}(u) = \hat{b}(u) \hat{a}(u)$$

Basic concept of CTRW model

$f_n(t|m)dt$:

Probability that the random walker starts at a site m at time 0 and arrives at a site n between time $t \sim t + dt$

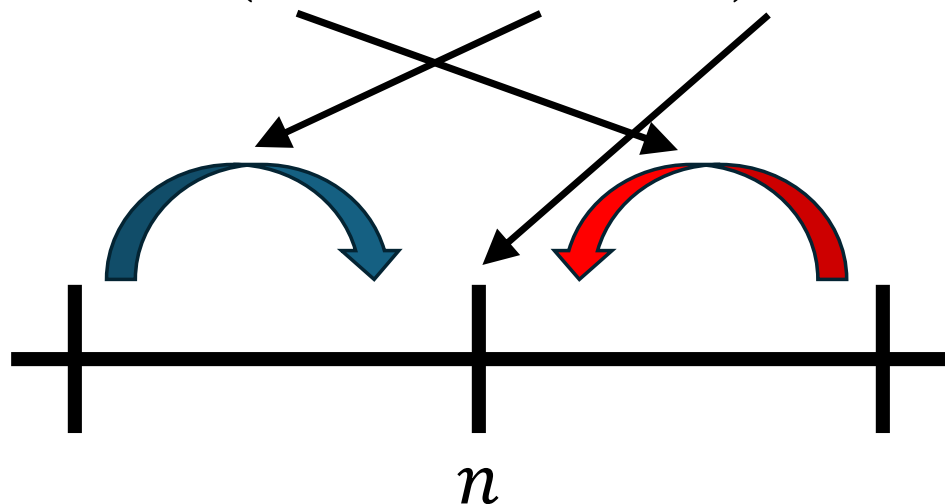
$$f_n(t|m) = \left(\int_0^t d\tau \Psi^R(\tau) f_{n-1}(t-\tau|m) + \int_0^t d\tau \Psi^L(\tau) f_{n+1}(t-\tau|m) \right) + \delta_{nm} \delta(t)$$

$$\hat{f}_n(u|m) \equiv \int_0^\infty dt e^{-u} f_n(t|m) = \left(\hat{\Psi}^R(u) \hat{f}_{n-1}(u|m) + \hat{\Psi}^L(u) \hat{f}_{n+1}(u|m) \right) + \delta_{nm}$$

$\psi^L(t)$ 와 $\psi^R(t)$ 가
같은 distribution을 따른다면.

$$\hat{\Psi}^L(0) = L \text{ \& } \hat{\Psi}^R(0) = R$$

$$= \hat{\Psi}(u) \left(R \hat{f}_{n-1}(u|m) + L \hat{f}_{n+1}(u|m) \right) + \delta_{nm}$$



1. Lattice site $n-1$ 에 시간 τ 에 도착한 random walker가 $t-\tau$ 의 시간 이후 오른쪽으로 움직여 lattice site n 에 도달하는 경우
2. Lattice site $n+1$ 에 시간 τ 에 도착한 random walker가 $t-\tau$ 의 시간 이후 왼쪽으로 움직여 lattice site n 에 도달하는 경우
3. 시간 0에 lattice site n 에 random walker가 생성된 경우. 이 경우에는 공간에 생성되는 순간이 도착하는 순간이 되어버린다.

Basic concept of CTRW model

$p_n(t|m)$:

Probability that the random walker starts at a site m at time 0 and is found at a site n at time t .

$$\hat{p}_n(u|m) = \frac{1 - \hat{\Psi}(u)}{u} \hat{f}_n(u|m) \quad \mathcal{L}^{-1} \left[\frac{1 - \hat{\Psi}(u)}{u} \right]: \text{Survival probability}$$

$$\hat{f}_n(u|m) = \hat{\Psi}(u) \left(R \hat{f}_{n-1}(u|m) + L \hat{f}_{n+1}(u|m) \right) + \delta_{nm}$$

$$\rightarrow \hat{p}_n(u|m) = \hat{D}(u) \left(R \hat{p}_{n-1}(u|m) + L \hat{p}_{n+1}(u|m) - \hat{p}_n(u|m) \right)$$

: **G**eneralized **M**aster **E**quation (GME) of random walker.

$$\hat{D}(u) = \frac{u \hat{\Psi}(u)}{1 - \hat{\Psi}(u)} : \text{discrete space Diffusion Kernel}$$

$$\hat{p}(n, u|m) = u \hat{p}(n, u|m) - \delta_{n,m}$$

$$\left(\because \int_0^\infty dt e^{-ut} \dot{a}(t) = (a(t)e^{-ut}) \Big|_{t=0}^\infty + u \int_0^\infty dt e^{-ut} a(t) = -a(0) + u \hat{a}(u) \right)$$

Basic concept of CTRW model

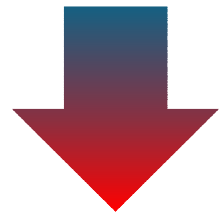
$$\hat{p}_n(u|m) = \frac{1 - \hat{\Psi}(u)}{u} \hat{f}_n(u|m) \quad \rightarrow \quad u\hat{p}_n(u|m) - \delta_{n,m} = (1 - \hat{\Psi}(u)) \hat{f}_n(u|m) - \delta_{n,m} = \hat{p}_n(u|m)$$

$$(1 - \hat{\Psi}(u)) \hat{f}_n(u|m) - \delta_{n,m} = (\hat{f}_n(u|m) - \delta_{n,m}) - \hat{\Psi}(u) \hat{f}_n(u|m) \quad \hat{f}_n(u|m) = \hat{\Psi}(u) (R\hat{f}_{n-1}(u|m) + L\hat{f}_{n+1}(u|m)) + \delta_{nm}$$

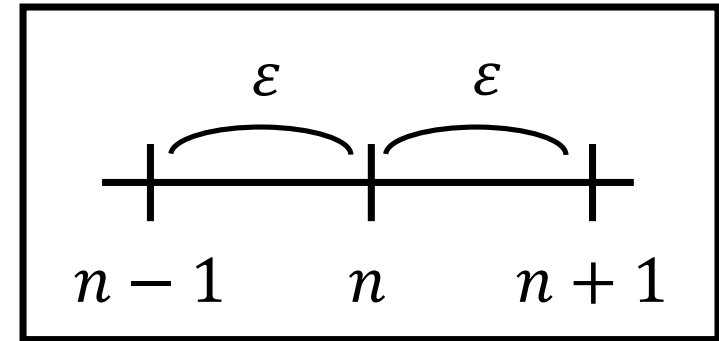
$$\begin{aligned} \hat{p}_n(u|m) &= \frac{1 - \hat{\Psi}(u)}{u} \hat{f}_n(u|m) \\ &= \hat{\Psi}(u) (R\hat{f}_{n-1}(u|m) + L\hat{f}_{n+1}(u|m) - \hat{f}_n(u|m)) \\ &= \frac{u\hat{\Psi}(u)}{1 - \hat{\Psi}(u)} (R\hat{p}_{n-1}(u|m) + L\hat{p}_{n+1}(u|m) - \hat{p}_n(u|m)) \\ &= \hat{D}(u) (R\hat{p}_{n-1}(u|m) + L\hat{p}_{n+1}(u|m) - \hat{p}_n(u|m)) \end{aligned} \quad \hat{D}(u) = \frac{u\hat{\Psi}(u)}{1 - \hat{\Psi}(u)}$$

Discrete to Continuous

$$\hat{p}_n(u|m) = \hat{D}(u)(R\hat{p}_{n-1}(u|m) + L\hat{p}_{n+1}(u|m) - \hat{p}_n(u|m))$$



$$\lim \varepsilon \rightarrow 0$$



$$\hat{P}(x, u|x_0) = \hat{D}(u)(\partial_x^2 + \beta U'(x)\partial_x)\hat{P}(x, u|x_0)$$

$$\hat{D}(u) = \frac{\varepsilon^2}{2}\hat{D}(u) \quad \beta U'(x) = 2(L - R)/\varepsilon$$

$$\hat{p}_n(u|m) = \varepsilon \hat{P}(x, u|x_0) \quad x = \varepsilon n, \quad x_0 = \varepsilon m$$

Diffusion Kernel value needs to be **invariant** to lattice constant ε .

$$\hat{D}(u) = \frac{\varepsilon^2}{2}\hat{D}(u): \text{diffusion kernel}$$

Discrete to Continuous

$$\hat{p}_n(u|m) = \hat{D}(u)(R\hat{p}_{n-1}(u|m) + L\hat{p}_{n+1}(u|m) - \hat{p}_n(u|m))$$

$$\hat{p}_n(u|m) = \varepsilon \hat{P}(x, u|x_0) \quad x = \varepsilon n, \quad x_0 = \varepsilon m$$

$$\hat{P}(x, u|x_0)$$

$$= (R + L)\hat{P}(x, u|x_0)$$

$$\begin{aligned} \hat{P}(x, u|x_0) &= \hat{D}(u) \left(R\hat{P}(x - \varepsilon, u|x_0) + L\hat{P}(x + \varepsilon, u|x_0) - \hat{P}(x, u|x_0) \right) \\ &= \hat{D}(u) \left(R\hat{P}(x - \varepsilon, u|x_0) + L\hat{P}(x + \varepsilon, u|x_0) - R\hat{P}(x, u|x_0) - L\hat{P}(x, u|x_0) \right) \end{aligned}$$

Taylor expansion

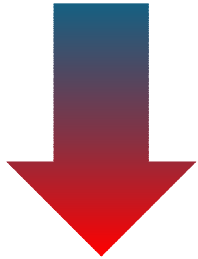
$$\begin{aligned} \hat{P}(x + \varepsilon, u|x_0) &\cong \hat{P}(x, u|x_0) + \varepsilon \partial_x \hat{P}(x, u|x_0) + \varepsilon^2/2 \partial_x^2 \hat{P}(x, u|x_0) + \dots \\ \hat{P}(x - \varepsilon, u|x_0) &\cong \hat{P}(x, u|x_0) - \varepsilon \partial_x \hat{P}(x, u|x_0) + \varepsilon^2/2 \partial_x^2 \hat{P}(x, u|x_0) + \dots \end{aligned}$$

$$\begin{aligned} \hat{P}(x, u|x_0) &= \hat{D}(u) \left(\varepsilon (L - R) \partial_x \hat{P}(x, u|x_0) + (L + R) \varepsilon^2/2 \partial_x^2 \hat{P}(x, u|x_0) \right) \\ &= \varepsilon^2 \hat{D}(u)/2 \left(2(L - R)/\varepsilon \times \partial_x \hat{P}(x, u|x_0) + (L + R) \partial_x^2 \hat{P}(x, u|x_0) \right) \end{aligned}$$

Discrete to Continuous

$$\begin{aligned}
 \hat{P}(x, u|x_0) &= \hat{D}(u) \left(\varepsilon (L - R) \partial_x \hat{P}(x, u|x_0) + (L + R) \varepsilon^2 / 2 \partial_x^2 \hat{P}(x, u|x_0) \right) \\
 &= \varepsilon^2 \hat{D}(u) / 2 \left(2(L - R) / \varepsilon \times \partial_x \hat{P}(x, u|x_0) + (L + R) \partial_x^2 \hat{P}(x, u|x_0) \right) \\
 &= \varepsilon^2 \hat{D}(u) / 2 \left(\color{red}{2(L - R)} / \varepsilon \times \partial_x \hat{P}(x, u|x_0) + \partial_x^2 \hat{P}(x, u|x_0) \right)
 \end{aligned}$$

(R + L) = 1



When $t \rightarrow \infty$,

$P(x, t|x_0)$ becomes equilibrium and follows **Boltzmann distribution**, i.e., $P_{\text{eq}}(x) \sim e^{-\beta U(x)}$

Tau-Berian theorem: $\lim_{u \rightarrow 0} u \hat{f}(u) = \lim_{t \rightarrow \infty} f(t)$

$$\begin{aligned}
 \lim_{u \rightarrow 0} u \hat{P}(x, u|x_0) &= \lim_{u \rightarrow 0} u \left(u \hat{P}(x, u|x_0) - \delta(x - x_0) \right) = \lim_{u \rightarrow 0} u \left(P_{\text{eq}}(x) - \delta(x - x_0) \right) = 0 \\
 &= \lim_{u \rightarrow 0} \left[\varepsilon^2 \hat{D}(u) / 2 \times \partial_x \left(2(L - R) / \varepsilon \times u \hat{P}(x, u|x_0) + \partial_x u \hat{P}(x, u|x_0) \right) \right]
 \end{aligned}$$

Discrete to Continuous

$$\begin{aligned}\lim_{u \rightarrow 0} u \hat{P}(x, u | x_0) &= \lim_{u \rightarrow 0} u \left(u \hat{P}(x, u | x_0) - \delta(x - x_0) \right) = \lim_{u \rightarrow 0} u \left(P_{\text{eq}}(x) - \delta(x - x_0) \right) = 0 \\ &= \lim_{u \rightarrow 0} \left[\varepsilon^2 \hat{D}(u)/2 \times \partial_x \left(2(L - R)/\varepsilon \times u \hat{P}(x, u | x_0) + \partial_x u \hat{P}(x, u | x_0) \right) \right]\end{aligned}$$

$$(1) \quad \lim_{u \rightarrow 0} \varepsilon^2 \hat{D}(u)/2 = \lim_{u \rightarrow 0} \hat{D}(u) \equiv \text{Constant}$$

$$(2) \quad \lim_{u \rightarrow 0} \left[2(L - R)/\varepsilon \times u \hat{P}(x, u | x_0) + \partial_x u \hat{P}(x, u | x_0) \right] = \underbrace{2(L - R)/\varepsilon \times P_{\text{eq}}(x) + \partial_x P_{\text{eq}}(x)}_{\text{Constant}}$$

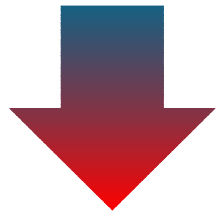
$$\alpha y(x) + y'(x) = \beta \quad \rightarrow [\alpha y(x) + y'(x)]e^{\alpha x} = [y(x)e^{\alpha x}]' = \beta e^{\alpha x}$$

$$\rightarrow y(x)e^{\alpha x} - y(0) = \int_0^x \beta e^{\alpha x} = \frac{\beta}{\alpha} [e^{\alpha x} - 1] \quad \rightarrow y(x) = \frac{\beta}{\alpha} + \left[y(0) - \frac{\beta}{\alpha} \right] e^{-\alpha x}$$

$$P_{\text{eq}}(x) \sim e^{-\beta U(x)}: \text{ Boltzmann distribution } \rightarrow \beta = 0, \alpha = 2(L - R)/\varepsilon = \beta U'(x)$$

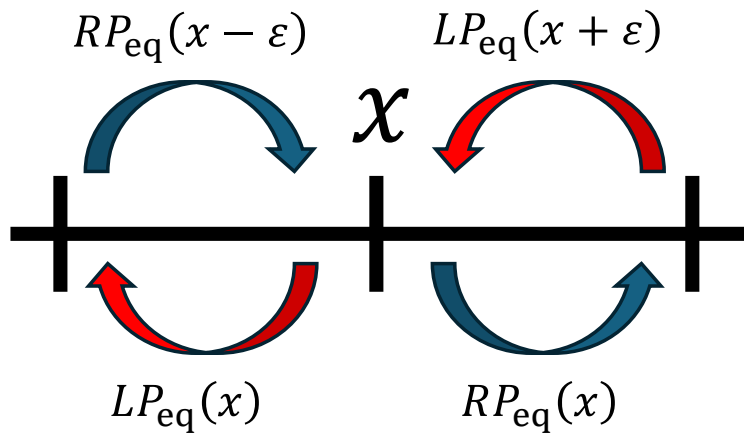
Discrete to Continuous

$$u\hat{P}(x, u|x_0) = u\hat{D}(u) \left(R\hat{P}(x - \varepsilon, u|x_0) + L\hat{P}(x + \varepsilon, u|x_0) - R\hat{P}(x, u|x_0) - L\hat{P}(x, u|x_0) \right)$$



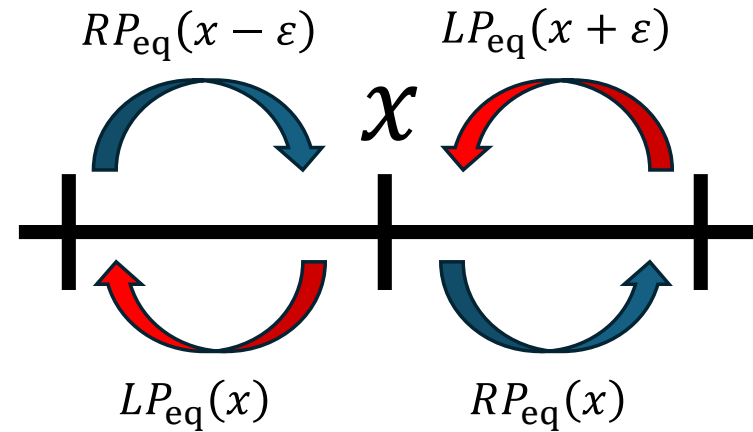
$$\lim u \rightarrow 0$$

$$\lim_{t \rightarrow \infty} \dot{P}(x, t|x_0) = 0 = u\hat{D}(0) \left(RP_{\text{eq}}(x - \varepsilon) + LP_{\text{eq}}(x + \varepsilon) - RP_{\text{eq}}(x) - LP_{\text{eq}}(x) \right)$$



들어온 만큼, 나가는 상황이 되기에, equilibrium에 도달한다.

Discrete to Continuous



$$\begin{aligned} \frac{L - R}{\varepsilon} &= \frac{LP_{eq}(x) - \textcolor{red}{RP}_{eq}(x)}{\varepsilon P_{eq}(x)} = \frac{LP_{eq}(x) - \textcolor{red}{LP}_{eq}(x + \varepsilon)}{\varepsilon P_{eq}(x)} = -\frac{P_{eq}(x + \varepsilon) - P_{eq}(x)}{\varepsilon} \times \frac{L}{P_{eq}(x)} = \frac{LP'_{eq}(x)}{P_{eq}(x)} \\ + \left[\frac{L - R}{\varepsilon} &= \frac{\textcolor{red}{LP}_{eq}(x) - RP_{eq}(x)}{\varepsilon P_{eq}(x)} = \frac{\textcolor{red}{RP}_{eq}(x - \varepsilon) - RP_{eq}(x)}{\varepsilon P_{eq}(x)} = -\frac{P_{eq}(x) - P_{eq}(x - \varepsilon)}{\varepsilon} \times \frac{R}{P_{eq}(x)} = \frac{RP'_{eq}(x)}{P_{eq}(x)} \right] \\ 2 \frac{L - R}{\varepsilon} &= -\frac{P'_{eq}(x)}{P_{eq}(x)} \rightarrow P_{eq}(x) \sim \exp\left(-\frac{2(L - R)}{\varepsilon} x\right) \end{aligned}$$

$$2(L - R)/\varepsilon = \beta U'(x) \quad (\because P_{eq}(x) \sim e^{-\beta U(x)})$$

Discrete to Continuous

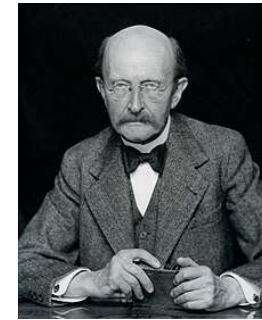
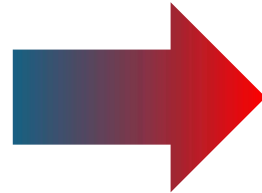
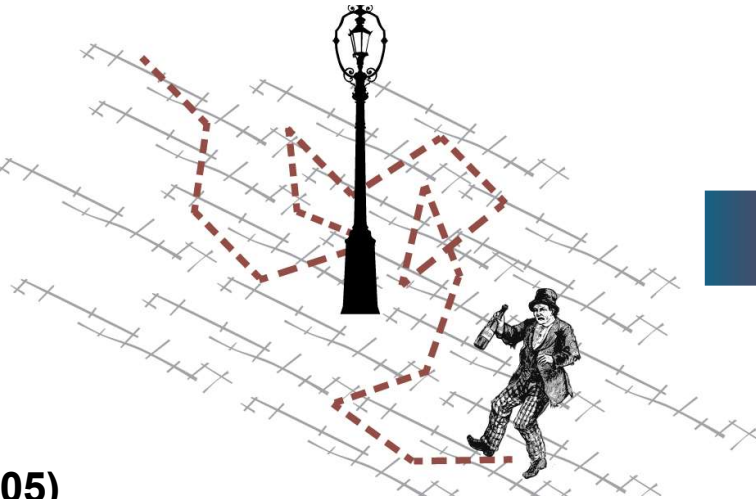
$$\hat{P}(x, u|x_0) = \varepsilon^2 \hat{D}(u)/2 \left(2(L - R)/\varepsilon \times \partial_x \hat{P}(x, u|x_0) + \partial_x^2 \hat{P}(x, u|x_0) \right)$$

$$= \hat{D}(u)(\beta U'(x)\partial_x + \partial_x^2)\hat{P}(x, u|x_0)$$

Fokker-Planck Operator



Karl Pearson (1905)



**Adriaan Fokker (1914), Max Planck (1917)
and Marian Smoluchowski (1916)**

- $\partial_t P(x, t) = [D(\partial_x^2 + \partial_x(\beta U)\partial_x)]P(x, t)$